

CSS Box Model Documentation

1. Box Model

The CSS box model is a fundamental concept that describes how elements are structured on a webpage. It consists of four main parts:

- **Content:** The actual text or image inside the element.
- **Padding:** Space between the content and the border.
- **Border:** The edge surrounding the padding (optional).
- **Margin:** Space between the element and surrounding elements.

Example:

```
.box {  
  width: 200px;  
  padding: 10px;  
  border: 5px solid black;  
  margin: 15px;  
}
```

2. Padding

Padding controls the space inside an element, between the content and the border.

Values:

- **Single value:** padding: 10px; (applies to all sides)
- **Two values:** padding: 10px 20px; (top-bottom, left-right)
- **Three values:** padding: 10px 15px 20px; (top, left-right, bottom)
- **Four values:** padding: 10px 15px 20px 25px; (top, right, bottom, left)

3. Margin

Margin controls the space outside an element, creating gaps between elements.

Values:

Similar to padding, margins can have 1 to 4 values.

- `margin: 10px;` (all sides)
- `margin: 10px 20px;` (top-bottom, left-right)
- `margin: 10px auto;` (top-bottom set, left-right centred)

Margin Collapse:

If two elements have margins, the larger margin is applied instead of adding both.

4. Border

Borders define the edge of an element.

Syntax:

```
border: width style color;
```

Example:

```
border: 2px solid red;
```

Border Styles:

- `solid`: A continuous line
- `dashed`: A broken line

- **dotted**: A series of dots
- **double**: Two solid lines

Border Sides:

You can set borders individually:

```
border-top: 2px solid black;  
border-right: 3px dashed blue;
```

5. Box Sizing

By default, the width and height apply only to the content, excluding padding and border. You can control this using `box-sizing`.

Values:

- **content-box** (default): Width and height exclude padding & border.
- **border-box**: Width and height include padding & border.

```
.box {  
  width: 200px;  
  padding: 10px;  
  border: 5px solid black;  
  box-sizing: border-box;  
}
```

6. Border Collapse

`border-collapse` is used to control the behavior of table borders.

Values:

- **separate** (default): Borders of adjacent table cells remain distinct.

- **collapse:** Adjacent table cell borders merge into one.

Example:

```
table {  
  border-collapse: collapse;  
  border: 1px solid black;  
}
```